David Calvert

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Summary: Ph.D. in computational astrophysics seeking a full-time position. Skilled in independent software development and 3D modeling with an emphasis on **MHD, plasma dynamics, data analytics, C++, Python, MATLAB, and FORTRAN**

Experience

Supernova Physicist 2019-2025

UC Berkeley | NC State University Berkeley, CA | Raleigh, NC

- Utilized a high-fidelity **MHD FORTRAN 3D supernova code** on **distributed HPC systems** to explore the role of turbulent plasma flow in supernova evolution
- Developed and implemented Python scripts to perform **data analysis and generate plots** used in publications
- Developed expertise in **MHD and plasma dynamics** through literature review and modeling experience
- Adapted numerical solvers for **nonlinear PDEs** with constraint preserving methods to enforce magnetic field divergence-free boundary conditions ($\vec{\nabla} \cdot \vec{B} = 0$) on a 3D cartesian grid

App Developer (Independent Contract) 2025

Kondra Chemical San Francisco, CA

- Designed, built, tested, and delivered a LabView app for measurement of chemical concentration in water samples
- Developed hardware control, spectrometer data acquisition, and concentration interpolation modules

Software Engineer 2024 - Present

KOINSLOT Inc. - GitHub San Francisco, CA

- Contributed to *Kywy*, a handheld "low-code" educational coding device
- Developed complete games, custom rotation algorithms for sprite rendering, and asset-generation scripts in **C++**

Business Systems Analyst 2018 - 2019

Insightsoftware Raleigh, NC

- Planned and executed data migrations for multiple acquired companies on short timescales
- Leveraged **SQL** and SOQL to perform data migrations into Salesforce from previous CRM systems

IT Systems Administrator Jan 2015 - May 2018

North Carolina State University Raleigh, NC

- Co-managed IT operations for the Physics Department at NC State University as one of two team members
- Implemented test environments in **Docker** and gained fluency in **Linux**

Education

Ph.D., Astrophysics Dec. 2025

UC Berkeley | North Carolina State University Berkeley, CA & Raleigh, NC

Dissertation: ***Magnetohydrodynamics of 3D Core-Collapse Supernovae***

B.S., Physics with Honors & Applied Mathematics May 2018

North Carolina State University Raleigh, NC

Publications

- Calvert, D., et al. (2025). "Turbulence in Core-Collapse Supernovae", *Astrophysical Journal*
- Calvert, D., et al. (2025). "The Interplay of Magnetic Fields and Vorticity in Core-Collapse Supernovae", *Astrophysical Journal*. In prep

Skills

Languages/Tools: Python (numpy, scipy, mpi4py), Docker, FORTRAN, C++, Git, MPI, OpenMP, OpenACC, VS Code, CUDA, Bash, MATLAB, LabView, Autodesk Fusion

Technical/Scientific: Plasma Physics, PDEs/ODEs, Vector Calculus, Numerical Methods, Multiphysics Simulation, Large Data Analytics, HPC Optimization, Statistical Analysis, Data Visualization

Soft Skills: Scientific Communication, Mentoring, Collaboration, Project Organization, Technical Document Review, Language Learning (German, Spanish, Hindi)